VEDANT KALE

Pune, India | +91 9356608540 | kalevedant750@gmail.com
linkedin.com/in/vedant-kale-developer | github.com/VedantKale244 | leetcode.com/u/Vedant_Kale27

PROFESSIONAL SUMMARY

Final-year B.Tech. CS student and Software Developer Intern targeting Software Development Engineer (SDE) roles in backend engineering and AI systems. Built RAG and GraphRAG pipelines that raised Recall@10 from 35.3% to 84.1%. Skilled in C++, Python, Java, and DSA; 150+ LeetCode problems solved.

EDUCATION

Pimpri Chinchwad University (PCU), Pune, India **Aug 2023 – May 2027**
B.Tech. in Computer Science & Engineering *CGPA: 7.69/10.0*

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript, TypeScript, SQL

Frameworks and Web: FastAPI, Node.js, Express.js, React.js, Next.js, Tailwind CSS, REST APIs, Microservices

Databases and Retrieval: PostgreSQL, MongoDB, Redis, Neo4j, Pinecone, pgvector, SQLAlchemy, BM25, Hybrid Search, Cross-Encoder Reranking

AI and ML: RAG, GraphRAG, Knowledge Graphs, Embeddings, RAGAS, LangChain, LangGraph, Ollama, Cohere Rerank, PyTorch, Scikit-learn, Pandas, NumPy

Cloud, DevOps and Testing: Docker, Git, Linux, CI/CD, AWS (EC2, S3, Lambda, DynamoDB), Pytest, Automated Testing, Latency Monitoring

Core CS: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Operating Systems, DBMS, System Design, Distributed Systems, Software Architecture

WORK EXPERIENCE

Software Developer Intern, Karst Technologies Private Limited, Pune, India **Jul 2026 – Sep 2026**
Technologies: JavaScript, Node.js, Express.js, React.js, MongoDB, REST APIs, Git

- Developed and maintained 3 application components, resolving 5 software defects across development workflows.
- Integrated and tested 4 REST API endpoints with database operations, validating data flow across components.
- Executed 10 functional test cases, documenting defects and verifying application behavior before feature delivery.

PROJECTS

Neuro-Adaptive GraphRAG Platform **Jan 2026 – Apr 2026**
Personal Project | Python, FastAPI, Neo4j, Pinecone, Cohere, RAGAS | github.com/VedantKale244/RAG-pipeline

- Built a hybrid GraphRAG pipeline (Neo4j, Pinecone, Cohere reranking) with 534 passing tests across 36 modules.
- Implemented RAGAS evaluation gates for retrieval quality and faithfulness with a 2.5 s p95 latency threshold.
- Developed a self-optimization framework (quality 50%, latency 30%, feedback 20%) with 15% canary rollout.

ArthaSEC – Auditable Financial RAG and Evidence Verification Engine **May 2026 – Jul 2026**
Personal Project | Python, FastAPI, BM25, Hybrid Search, SQLAlchemy | github.com/VedantKale244/ArthaSEC

- Built a financial retrieval engine over 352 SEC filings (31,962 chunks), improving Recall@10 from 35.3% to 84.1% (88.4% annual-only) on a 232-question benchmark using metadata scoping and content-focused BM25.
- Implemented citation-backed financial answers and deterministic growth calculations, validating company, period, and scale; passed 53 real-corpus acceptance checks and 14 constructed adversarial cases.
- Reduced fast-mode p50 latency by 72%, from 77.5 ms to 22.1 ms, via indexed metadata filters and precomputed BM25 scoring; engineered bounded FastAPI concurrency with 544 Python regression tests.

Adaptive Two-Tier Malware Detection **Ongoing Research**
Python, Machine Learning, Random Forest, Evidential Deep Learning, API Security *Research Project*

- Designed a two-tier detection framework (Random Forest + sandbox) for SQLi, XSS, malware uploads, and RCE on 9,000 API requests.
- Added confidence-based escalation and feedback retraining: 98.1% accuracy, 58.8% fewer deep inspections.
- Evaluated on 10,000 CIC-IDS/EMBER records, reaching 99.30% accuracy and 0.003 ms CPU inference latency.

CERTIFICATIONS

- Data Structures and Algorithms Specialization – Amazon (via Coursera), 2025
- The Complete Full-Stack Web Development Bootcamp – Udemy, 2026

- Software Design and Architecture Specialization – Coursera (In Progress)

ACHIEVEMENTS

- **Competitive Programming:** LeetCode rating 1,480 (150+ solved); CodeChef rating 1,824; CodeChef DSA rating 1,659.